FLOWSENSE

Week 41 queue review

SCOPE
2026-10-05 to 2026-10-09, all checkpoints

GENERATED BY
Administrator

GENERATED AT
2026-10-12 13:51 (local airport time)

CONTENTS

Key findings · KPI summary · By checkpoint · By hour of day · Daily trend · Wait time distribution · Flights and counters · SLA breach details · Staffing recommendations · Your sections (3) · Glossary

All averages are passenger-weighted: SUM(value x pax) / SUM(pax). SLA checks use each checkpoint's current threshold. Values marked (!) in bold are above the SLA.

Key findings

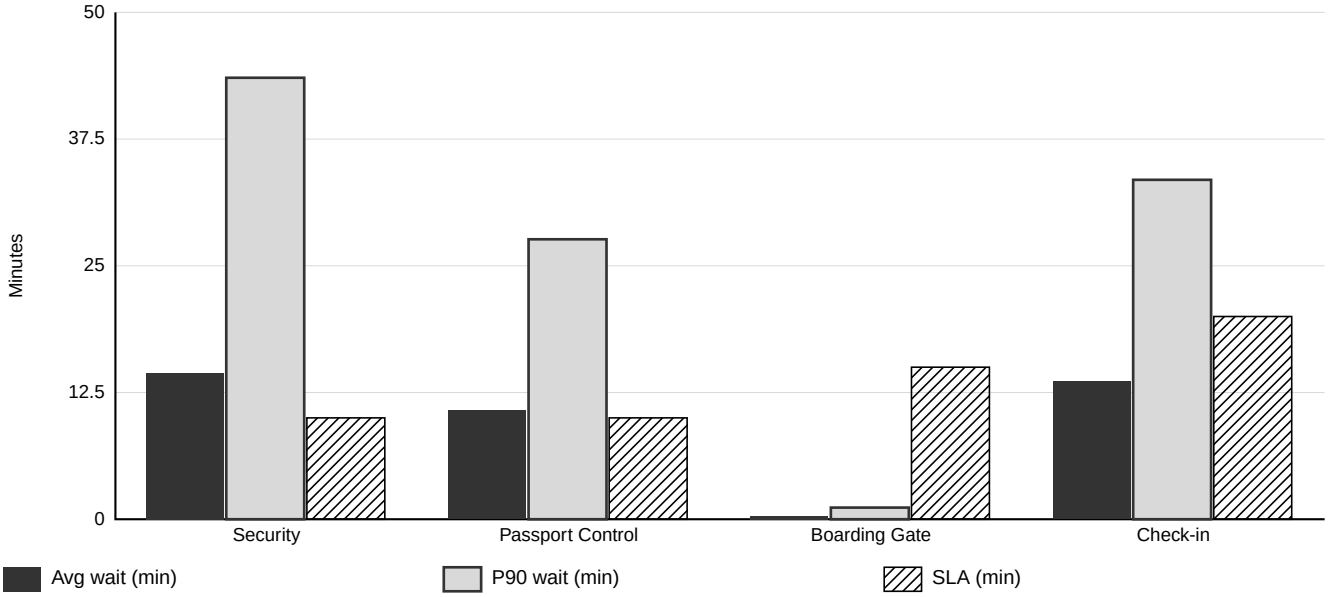
- 9575 passengers in 5318 groups were observed. They waited 10.5 min on average (median 2.2 min, 9 in 10 under 29.7 min) and 69.2 % were served within the SLA.
- 2949 passengers (1616 groups) waited longer than their checkpoint's SLA.
- Average wait above the SLA at Security (14.4 min vs 10 min), Passport Control (10.7 min vs 10 min).
- Lowest SLA compliance: Security, 56.4 %.
- Busiest hour: 09:00 with 2501 passengers (avg wait 20.1 min).
- Worst day: 2026-10-09, avg wait 22.7 min, SLA compliance 49.1 %.
- At the busiest hour, more counters are needed than were open at Security (3 open, 4 needed at 09:00), Passport Control (3 open, 4 needed at 08:00), Check-in (6 open, 9 needed at 09:00).
- Flight with the most SLA breaches: LH1312, 397 passengers (65.0 % compliance).

KPI summary

Indicator	Value	What it means
Passengers	9575	People observed (a group of 4 counts 4).
Groups	5318	Observation rows (families or travel groups).
Avg wait (min)	10.5	Queue join to counter, passenger-weighted.
P50 wait (min)	2.2	Half of the passengers waited less than this.
P90 wait (min)	29.7	9 in 10 passengers waited less than this.
Avg processing (min)	4.1	Time at the counter.
Avg total time (min)	14.6	Wait plus processing.
Throughput (pax/h)	67.9	Passengers served per observed hour.
SLA compliance	**(!) 69.2 %**	Share of passengers within the SLA.
SLA breaches	**(!) 2949 pax in 1616 groups**	Passengers who waited longer than the SLA.

By checkpoint

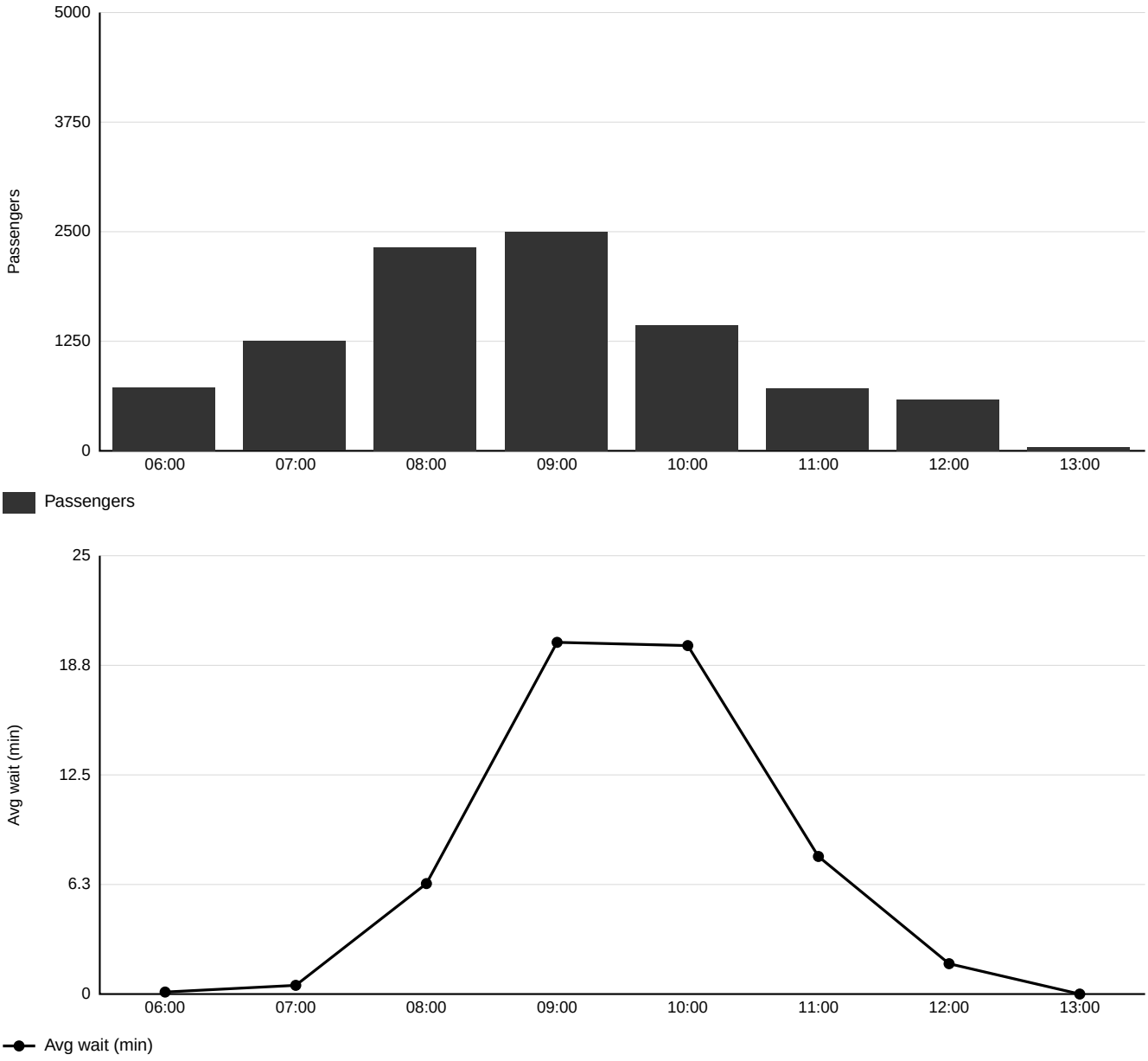
Average wait against each checkpoint's SLA target, then the detailed figures.



Checkpoint	Pax	Avg wait	P90 wait	Avg proc.	Pax/h	SLA	Compliance	Peak hour
Security	2578	**(!) 14.4**	43.5	3.6	73	10	56.4 %	09:00 (128/h)
Passport Control	2516	**(!) 10.7**	27.6	3.2	71	10	57.4 %	08:00 (132/h)
Boarding Gate	1829	0.3	1.1	1.6	53	15	100.0 %	09:00 (102/h)
Check-in	2652	13.6	33.5	6.9	74	20	71.6 %	09:00 (142/h)

Minutes unless stated. Pax/h: passengers served per observed hour. Peak hour: the hour with the most arrivals per day observed.

By hour of day

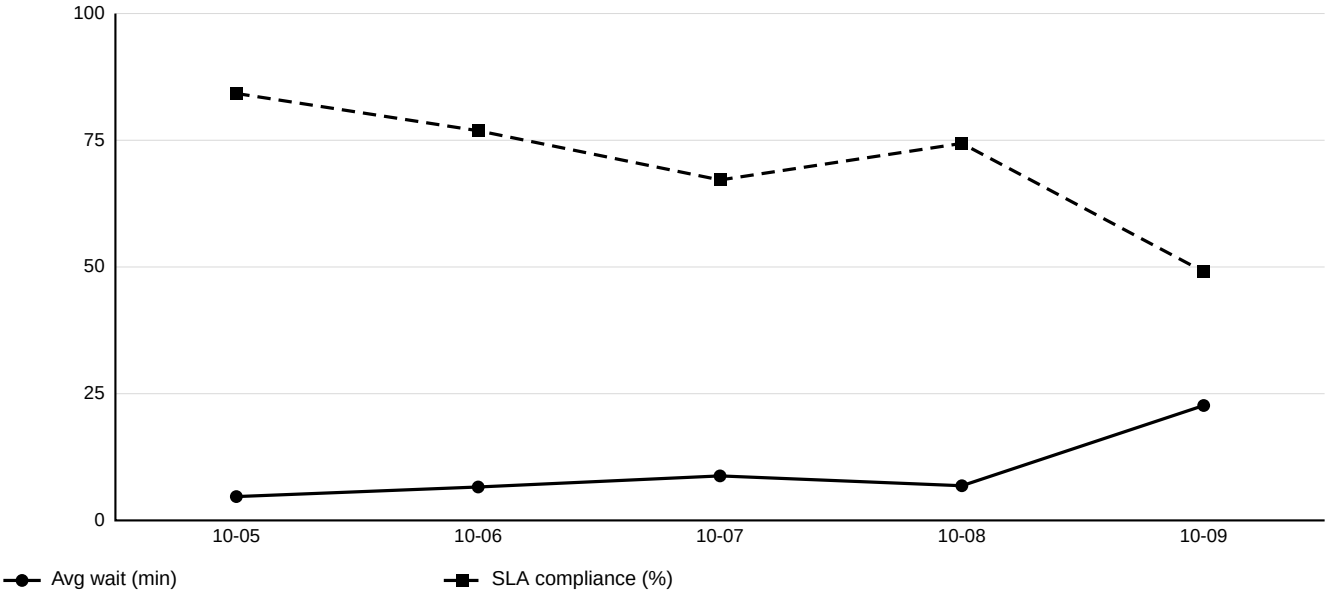


Hour	Passengers	Pax per day	Avg wait (min)	SLA compliance	Breach pax
06:00-06:59	721	144.2	0.1	100.0 %	0
07:00-07:59	1252	250.4	0.5	100.0 %	0
08:00-08:59	2323	464.6	6.3	82.8 %	**(!) 400**
09:00-09:59	2501	500.2	20.1	38.9 %	**(!) 1527**
10:00-10:59	1436	287.2	19.9	40.7 %	**(!) 851**
11:00-11:59	709	141.8	7.8	80.8 %	**(!) 136**
12:00-12:59	589	117.8	1.7	94.1 %	**(!) 35**
13:00-13:59	44	8.8	0.0	100.0 %	0

Avg wait (min) by hour and checkpoint

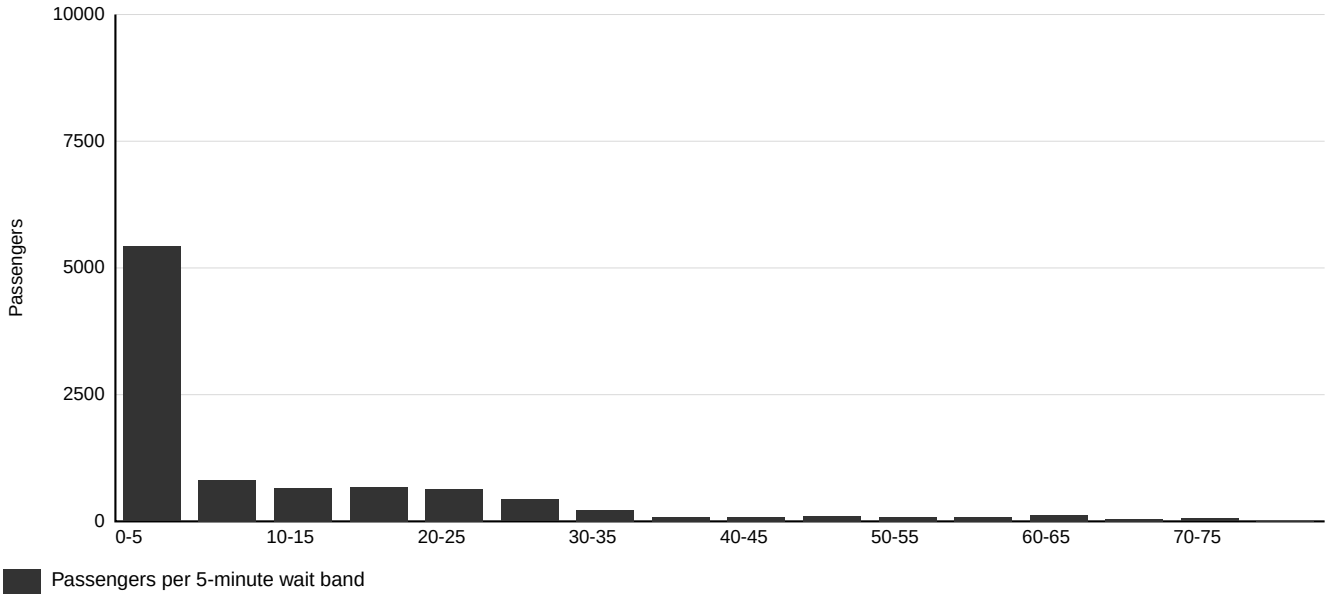
Hour	Boarding Gate	Check-in	Passport Control	Security
06:00	0.0	0.0	0.3	0.1
07:00	0.2	0.9	0.5	0.3
08:00	0.4	7.1	7.3	8.7
09:00	0.5	**(!) 26.3**	**(!) 22.0**	**(!) 26.8**
10:00	0.3	**(!) 26.5**	**(!) 19.2**	**(!) 26.8**
11:00	0.0	12.2	2.7	**(!) 12.4**
12:00	0.0	0.9	0.0	5.0
13:00	0.0	0.0	0.0	0.0

Daily trend



Date	Passengers	Avg wait (min)	SLA compliance	Breach pax
2026-10-05	1668	4.7	84.2 %	263
2026-10-06	1795	6.6	76.9 %	415
2026-10-07	1928	8.8	67.2 %	633
2026-10-08	1948	6.8	74.4 %	499
2026-10-09	2236	22.7	49.1 %	1139

Wait time distribution

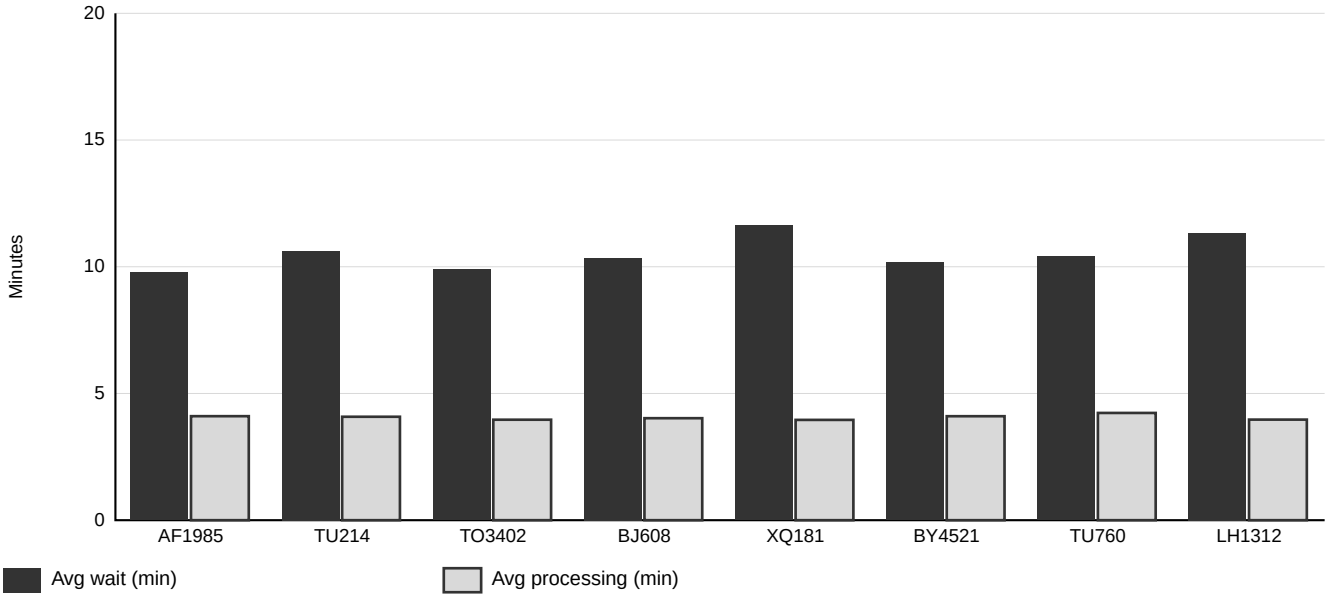


Median wait 2.2 min; 9 in 10 passengers waited under 29.7 min. 65.1 % waited 10 min or less; 71.9 % waited 15 min or less; 79.0 % waited 20 min or less.

Wait (min)	Passengers	Groups	Share	Cumulative
0-5	5424	3020	56.6 %	56.6 %
5-10	809	455	8.4 %	65.1 %
10-15	655	368	6.8 %	71.9 %
15-20	679	364	7.1 %	79.0 %
20-25	644	365	6.7 %	85.8 %
25-30	433	222	4.5 %	90.3 %
30-35	232	135	2.4 %	92.7 %
35-40	93	53	1.0 %	93.7 %
40-45	89	52	0.9 %	94.6 %
45-50	106	65	1.1 %	95.7 %
50-55	87	45	0.9 %	96.6 %
55-60	80	45	0.8 %	97.5 %
60-65	128	69	1.3 %	98.8 %
65-70	47	21	0.5 %	99.3 %
70-75	57	32	0.6 %	99.9 %
75-80	12	7	0.1 %	100.0 %

Flights and counters

By flight



Flight	Passengers	Groups	Avg wait (min)	Avg proc. (min)	SLA compliance	Breach pax
AF1985	1251	672	9.8	4.1	70.3 %	**(!) 371**
TU214	1239	691	10.6	4.1	71.3 %	**(!) 355**
TO3402	1238	691	9.9	4.0	70.0 %	**(!) 371**
BJ608	1211	676	10.3	4.0	68.2 %	**(!) 385**
XQ181	1196	683	11.6	4.0	67.2 %	**(!) 392**
BY4521	1169	642	10.2	4.1	69.1 %	**(!) 361**
TU760	1136	652	10.4	4.2	72.1 %	**(!) 317**
LH1312	1135	611	11.3	4.0	65.0 %	**(!) 397**

By counter

Counter	Passengers	Groups	Avg wait (min)	Avg proc. (min)	SLA compliance	Breach pax
G1	917	522	0.3	1.6	100.0 %	0
G2	912	508	0.3	1.7	100.0 %	0
S3	868	473	15.1	3.6	55.2 %	**(!) 389**
S2	857	464	13.6	3.7	57.5 %	**(!) 364**
S1	853	474	14.5	3.6	56.5 %	**(!) 371**
P1	843	475	11.0	3.2	56.0 %	**(!) 371**
P3	840	465	10.5	3.2	58.2 %	**(!) 351**
P2	833	469	10.6	3.3	58.1 %	**(!) 349**
C1	467	236	13.3	7.3	72.4 %	**(!) 129**
C3	449	252	13.4	6.9	71.0 %	**(!) 130**

Counter	Passengers	Groups	Avg wait (min)	Avg proc. (min)	SLA compliance	Breach pax
C4	449	240	13.4	6.9	71.5 %	**(!) 128**
C2	440	252	13.7	6.7	70.5 %	**(!) 130**
C5	436	247	13.2	6.9	72.9 %	**(!) 118**
C6	411	241	14.5	6.8	71.0 %	**(!) 119**

SLA breach details

By checkpoint

Checkpoint	SLA (min)	Passengers	Avg wait (min)	SLA compliance	Breach pax	Breach groups
Security	10	2578	**(!) 14.4**	56.4 %	**(!) 1124**	606
Passport Control	10	2516	**(!) 10.7**	57.4 %	**(!) 1071**	594
Boarding Gate	15	1829	0.3	100.0 %	0	0
Check-in	20	2652	13.6	71.6 %	**(!) 754**	416

Hours with the most breaches

Hour	Breach pax	Share of hour	Avg wait (min)
09:00	1527	61.1 %	20.1
10:00	851	59.3 %	19.9
08:00	400	17.2 %	6.3
11:00	136	19.2 %	7.8
12:00	35	5.9 %	1.7

Flights with the most breaches

Flight	Breach pax	Passengers	SLA compliance	Avg wait (min)
LH1312	397	1135	65.0 %	11.3
XQ181	392	1196	67.2 %	11.6
BJ608	385	1211	68.2 %	10.3
AF1985	371	1251	70.3 %	9.8
TO3402	371	1238	70.0 %	9.9
BY4521	361	1169	69.1 %	10.2
TU214	355	1239	71.3 %	10.6
TU760	317	1136	72.1 %	10.4

Staffing recommendations (M/M/c)

Checkpoint	lambda /h	mu /h	c	rho	Wq (min)	Status	Needed (avg)	Needed (peak)
Security	73.6	35.1	3	69.9 %	0.93	Within SLA	3	**4**

Checkpoint	lambda /h	mu /h	c	rho	Wq (min)	Status	Needed (avg)	Needed (peak)
Passport Control	72.0	37.3	3	64.4 %	0.62	Within SLA	3	**4**
Boarding Gate	52.9	74.1	2	35.7 %	0.12	Within SLA	1	2
Check-in	75.7	18.1	6	69.8 %	0.61	Within SLA	5	**9**

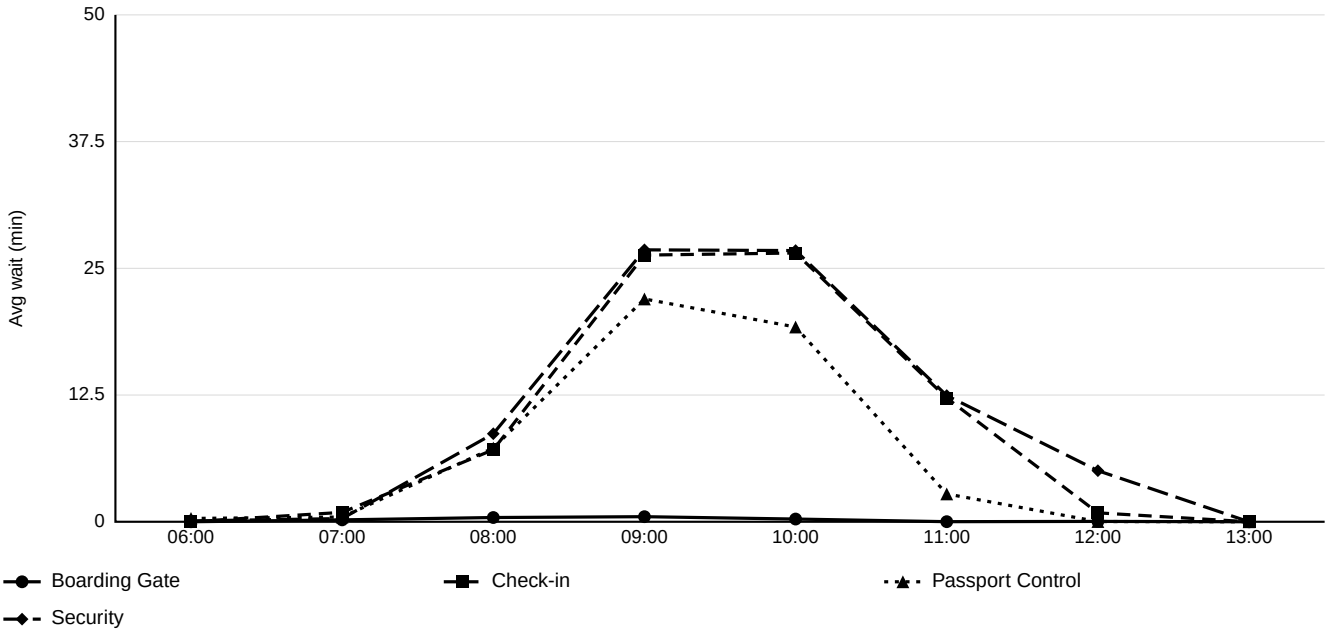
Reasoning

- Security: about 74 passengers arrive per hour and one counter serves about 35.1 per hour, so 3 counter(s) are busy 69.9 % of the time. Over the whole period the model predicts a 0.9 min wait against a 10 min SLA. Yet passengers actually waited 14.4 min on average: arrivals are not spread evenly, and the queue builds at peak times. At the busiest hour (09:00, about 128 passengers per hour) 4 counter(s) are needed to stay within the SLA. Recommendation: open 4 counters at peak times (1 more than observed).
- Passport Control: about 72 passengers arrive per hour and one counter serves about 37.3 per hour, so 3 counter(s) are busy 64.4 % of the time. Over the whole period the model predicts a 0.6 min wait against a 10 min SLA. Yet passengers actually waited 10.7 min on average: arrivals are not spread evenly, and the queue builds at peak times. At the busiest hour (08:00, about 132 passengers per hour) 4 counter(s) are needed to stay within the SLA. Recommendation: open 4 counters at peak times (1 more than observed).
- Boarding Gate: about 53 passengers arrive per hour and one counter serves about 74.1 per hour, so 2 counter(s) are busy 35.7 % of the time. Over the whole period the model predicts a 0.1 min wait against a 15 min SLA. At the busiest hour (09:00, about 102 passengers per hour) 2 counter(s) are needed to stay within the SLA. Recommendation: 2 counters are enough; 1 would do outside the peak.
- Check-in: about 76 passengers arrive per hour and one counter serves about 18.1 per hour, so 6 counter(s) are busy 69.8 % of the time. Over the whole period the model predicts a 0.6 min wait against a 20 min SLA. At the busiest hour (09:00, about 142 passengers per hour) 9 counter(s) are needed to stay within the SLA. Recommendation: open 9 counters at peak times (3 more than observed).

Needed (avg): fewest counters whose predicted wait is within the SLA at the period's average arrival rate. Needed (peak): the same at the busiest hour's arrival rate. The model assumes random arrivals and service times; real queues at peaks are usually longer.

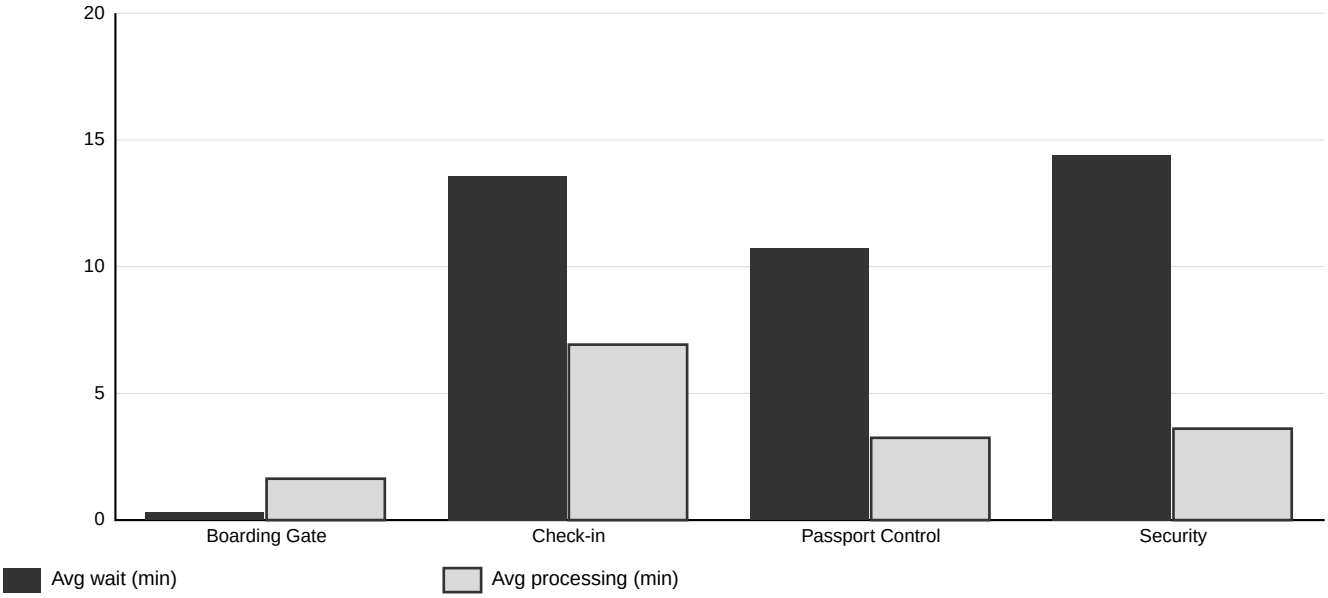
1. Avg wait by hour and checkpoint

Filters: none (all data).



2. Wait and processing by checkpoint

Filters: none (all data).



3. Shifts at a glance

Filters: none (all data).

Shift	Passengers (pax)	Avg wait (min)	SLA compliance (%)
Morning	9575	10.5	69.2 %

Glossary

Term	Meaning
lambda (arrival rate)	Passengers arriving per hour (per minute in the model).
mu (service rate)	Passengers one counter serves per hour: total passengers / total processing time.
c (counters)	Counters open at the same time: those recorded in the data, else the configured number.
rho (utilisation)	lambda / (c x mu): share of counter time spent serving. At 100 % or more the queue never clears.
Wq	Predicted wait in the queue from the M/M/c (Erlang C) model.
P50 / P90	Half / 9 in 10 of the passengers waited less than this.
SLA compliance	Share of passengers whose wait was within the checkpoint's SLA threshold.
Passenger-weighted	Each group counts as many times as it has passengers.

(Greek letters are spelled out because the PDF uses the standard Helvetica font.)